

Calculus 2

9 Lecture 9

n -Dimensional Riemann Integral: Part I

Definition 9.1. The set

$$I = \{\bar{x} \in \mathbb{R}^n \mid a_i \leq x_i \leq b_i, i = 1, 2, \dots, n\} = [a_1, b_1] \times [a_2, b_2] \times \dots \times [a_n, b_n], \quad (1)$$

where $\bar{x} = (x_1, \dots, x_n)$, $\bar{a} = (a_1, \dots, a_n)$, $\bar{b} = (b_1, \dots, b_n) \in \mathbb{R}^n$, is called a *nondegenerative n -dimensional closed interval*.

Definition 9.2. Let an interval I be defined by [Equality \(1\)](#). Then, the value

$$|I| \stackrel{\text{def}}{=} \prod_{i=1}^n (b_i - a_i)$$

is called the *measure* (or *n -measure* or *n -volume*) of the interval I .

Note that, since I is a nondegenerative interval (that is, $a_i < b_i$, $i = 1, 2, \dots, n$), we have $|I| > 0$.

Definition 9.3. A *partition* P of a one-dimensional nondegenerative interval $I = [a, b] \subset \mathbb{R}$ is a finite system of points $x_0, \dots, x_n$ of the interval such that

$$a = x_0 < x_1 < x_2 < \dots < x_n = b.$$

The intervals $[x_{i-1}, x_i]$, $i = 1, 2, \dots, n$, are called the intervals of the partition P . Note that

$$I = \bigcup_{i=1}^n [x_{i-1}, x_i].$$

Definition 9.4. Let an interval I be defined by [Equality \(1\)](#). Then, partitions of the coordinate intervals $[a_i, b_i]$, $i = 1, \dots, n$, induce a partition of the interval I into finer intervals obtained as the direct products of the intervals of the partitions of the coordinate intervals. More precisely:

Let $I = [a_1, b_1] \times [a_2, b_2] \times \dots \times [a_n, b_n]$ be defined by [Equality \(1\)](#). Then, partitions of the coordinate intervals $[a_i, b_i]$

$$\begin{aligned} P_1 : \quad & a_1 = x_{10} < x_{11} < x_{12} < \dots < x_{1t_1} = b_1, \\ P_2 : \quad & a_2 = x_{20} < x_{21} < x_{22} < \dots < x_{2t_2} = b_2, \\ & \vdots \\ P_n : \quad & a_n = x_{n0} < x_{n1} < x_{n2} < \dots < x_{nt_n} = b_n, \end{aligned}$$

induce a partition of I into $t_1 t_2 \dots t_n$ many intervals

$$I_{i_1 i_2 \dots i_n} = [x_{1(i_1-1)}, x_{1i_1}] \times [x_{2(i_2-1)}, x_{2i_2}] \times \dots \times [x_{n(i_n-1)}, x_{ni_n}] \subset \mathbb{R}^n.$$

The representation of the interval I as the union

$$I = \bigcup_{i_1=1}^{t_1} \bigcup_{i_2=1}^{t_2} \dots \bigcup_{i_n=1}^{t_n} I_{i_1 i_2 \dots i_n}$$

of finer intervals $I_{i_1 i_2 \dots i_n}$ just described will be called a *partition of the (n -dimensional) interval I* , and will be denoted by P .

Note that, all intervals $I_{i_1 i_2 \dots i_n}$ are nondegenerate by construction.

For example: Let $I = [a_1, b_1] \times [a_2, b_2]$ be a 2-dimensional interval. Then partitions of the coordinate intervals

$$P_1 : \quad a_1 = x_{10} < x_{11} < x_{12} < x_{13} = b_1,$$

(partition of $[a_1, b_1]$ into $t_1 = 3$ many intervals $[x_{1(i_1-1)}, x_{1i_1}]$), and

$$P_2 : \quad a_2 = x_{20} < x_{21} < x_{22} = b_2,$$

(partition of $[a_2, b_2]$ into $t_2 = 2$ many intervals $[x_{2(i_2-1)}, x_{2i_2}]$), induce a partition of I into $t_1 t_2 = 3 \cdot 2 = 6$ many intervals:

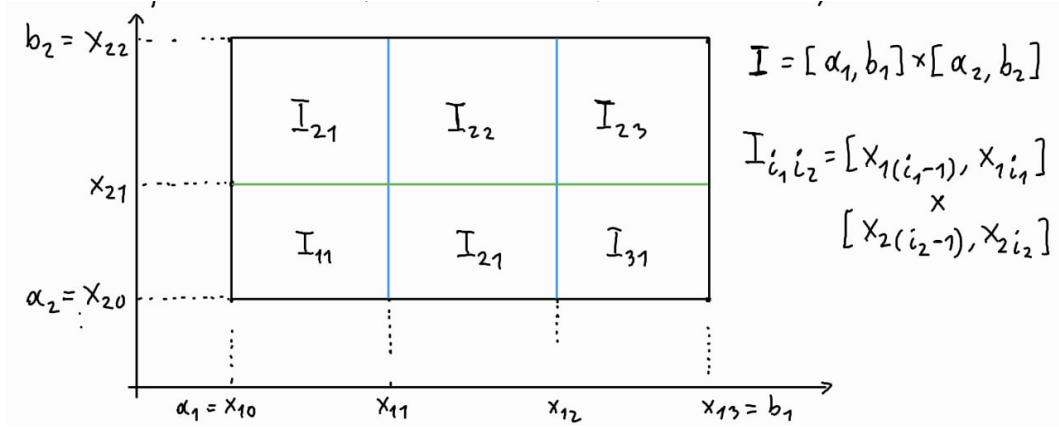


Figure 9.1

Definition 9.5. Let P be a partition of an interval I into finer intervals $I_{i_1 i_2 \dots i_n}$ (see Def 9.4). If in every interval $I_{i_1 i_2 \dots i_n}$ we fix a point $\zeta_{i_1 i_2 \dots i_n} \in I_{i_1 i_2 \dots i_n}$, then we say that we have a *tagged partition* (or a *partition with distinguished points*). The set of all $t_1 t_2 \dots t_n$ many points $\zeta_{i_1 i_2 \dots i_n}$ we will denote by the single letter ζ , and the tagged partition of I we will denote by (P, ζ) .

Definition 9.6. Let X be a non-empty subset of $\mathbb{R}^n$. Then, the *diameter* of X we will define as

$$d(X) = \sup_{\bar{a}, \bar{b} \in X} \|\bar{a} - \bar{b}\|,$$

where

$$\|\bar{a} - \bar{b}\| = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + \dots + (a_n - b_n)^2}$$

for $\bar{a} = (a_1, a_2, \dots, a_n)$, $\bar{b} = (b_1, b_2, \dots, b_n)$.

Note that, if an interval I is defined by Equality (1), then it ought to be clear that

$$d(I) = \|\bar{a} - \bar{b}\| > 0.$$

Definition 9.7. Let P be a partition of an interval I (see Def 9.4). Then, the maximum among the diameters of the intervals $I_{i_1 i_2 \dots i_n}$ of the partition P is called the *mesh* of the partition P and denoted by $\lambda(P)$. That is,

$$\lambda(P) = \max_{\substack{i_1 \in \{1, \dots, t_1\} \\ \vdots \\ i_n \in \{1, \dots, t_n\}}} d(I_{i_1 i_2 \dots i_n}).$$

Definition 9.8. Let

- I be an interval defined by Equality (1);
- $(P, \bar{\zeta})$ be a tagged partition of I (see Def 9.5);
- $|I_{i_1 i_2 \dots i_n}|$ be the measure of an interval $I_{i_1 i_2 \dots i_n}$ from partition $(P, \bar{\zeta})$ (see Def 9.2);

- $f : I \rightarrow \mathbb{R}$ be a function.

Then, the sum

$$\sigma(f, P, \bar{\zeta}) \stackrel{\text{def}}{=} \sum_{i_1=1}^{t_1} \sum_{i_2=1}^{t_2} \cdots \sum_{i_n=1}^{t_n} f(\zeta_{i_1 i_2 \dots i_n}) |I_{i_1 i_2 \dots i_n}|$$

is called the *Riemann sum* (of the function f corresponding to the tagged partition $(P, \bar{\zeta})$ of the interval I).

Definition 9.9. Let I be an interval defined by [Equality \(1\)](#) and let $f : I \rightarrow \mathbb{R}$ be a function. We say that the *(multiple) Riemann integral* of the function f over the interval I (denoted by $\int_I f(\bar{x}) d\bar{x}$) exists and equals s if for every $\varepsilon > 0$, there exists $\delta > 0$ such that for any tagged partition $(P, \bar{\zeta})$ of the interval I whose mesh (i.e. $\lambda(P)$) is less than δ , we have

$$|\sigma(f, P, \bar{\zeta}) - s| < \varepsilon,$$

where $\sigma(f, P, \bar{\zeta})$ is the Riemann sum (see [Def 9.8](#)). That is,

$$\left(\int_I f(\bar{x}) d\bar{x} = s \right) \iff \left(\lim_{\lambda(P) \rightarrow 0} \sigma(f, P, \bar{\zeta}) = s \right), \quad (2)$$

or, equivalently,

$$\forall \varepsilon > 0 \exists \delta(\varepsilon) > 0 \forall (P, \bar{\zeta}) \left[\lambda(P) < \delta \implies |\sigma(f, P, \bar{\zeta}) - s| < \varepsilon \right].$$

If limit (2) exists, we say that the function $f : I \rightarrow \mathbb{R}$ is *Riemann integrable over the interval I* .

The set of all Riemann integrable over the interval $I \subset \mathbb{R}^n$ functions we shall denote by $\mathcal{R}(I)$.

Equivalent but more expanded notation of the multiple integral is

$$\int_I f(\bar{x}) d\bar{x} = \int_I f(x_1, \dots, x_n) dx_1 \cdots dx_n = \int_I \cdots \int_I f(x_1, \dots, x_n) dx_1 \cdots dx_n.$$

If $n = 2$, then the integral is called the *double integral* and usually denoted by

$$\iint_I f(x, y) dx dy.$$

If $n = 3$, then the integral is called the *triple integral* and usually denoted by

$$\iiint_I f(x, y, z) dx dy dz.$$

Note that our process of constructing the multiple Riemann integral over an interval $I \subset \mathbb{R}^n$ is a verbatim repetition of the procedure of constructing the Riemann integral over a one-dimensional interval from Calculus-1.

Criteria for Riemann Integrability

Definition 9.10. A set $E \subset \mathbb{R}^n$ has *(n -dimensional) Lebesgue measure zero* if, for every $\varepsilon > 0$, there exist at most countable system $\{I_i\}$ of n -dimensional intervals I_i (see [Def 9.1](#)) such that

$$E \subseteq \bigcup_i I_i \quad \text{and} \quad \sum_i |I_i| \leq \varepsilon,$$

where $|I_i|$ is the measure of I_i (see [Def 9.2](#)).

If $E \subset \mathbb{R}^n$ is a set of Lebesgue measure zero, we denote this by $\mu(E) = 0$.

Note that, since the series $\sum |I_i|$ converges absolutely, by [Claim 3.9](#), the exact order of summation does not affect the sum, so that [Def 9.10](#) is unambiguous.

Claim 9.1 (Examples of sets of Lebesgue measure zero). 1. The union of a finite or countable number of sets of Lebesgue measure zero is a set of Lebesgue measure zero.

2. Any finite or countable set is a set of Lebesgue measure zero.

3. A subset of a set of Lebesgue measure zero is itself of Lebesgue measure zero.

4. There are examples of uncountable 1-dimensional sets of Lebesgue measure zero.

Proof. 1) Let $E = \bigcup_n E_n$ be at most countable union of sets $E_n \subset \mathbb{R}^n$ such that $\mu(E_n) = 0$ (see [Def 9.10](#)). Then, given $\varepsilon > 0$, for each E_n , by [Def 9.10](#), there exists at most countable system $\{I_{nk}\}$ of intervals $I_{nk} \subset \mathbb{R}^n$ such that

$$E_n \subseteq \bigcup_k I_{nk} \quad \text{and} \quad \sum_k |I_{nk}| < \frac{\varepsilon}{2^n}.$$

Since the union of at most countably many sets is itself at most countable, the system $\bigcup_n \{I_{nk}\}$ is at most countable. Since, for every n , $E_n \subseteq \bigcup_k I_{nk}$, we have

$$E = \bigcup_n E_n \subseteq \bigcup_n \bigcup_k I_{nk} = \bigcup I_{nk}.$$

Note that, by the definition of the union of sets, the order in which we unite the sets I_{nk} in $\bigcup I_{nk}$ is of no importance. Since for every n , $\sum_k |I_{nk}| < \varepsilon/2^n$, it should be clear that any finite sum of $|I_{nk}|$ is bounded by the value

$$\frac{\varepsilon}{2} + \frac{\varepsilon}{2^2} + \cdots = \varepsilon.$$

Therefore, we have

$$\sum_{n,k} |I_{nk}| \leq \varepsilon.$$

Note that, if the sum is countable, then the series $\sum_{n,k} |I_{nk}|$ converges absolutely (by [Claim 2.2](#)), and the value of $\sum_{n,k} |I_{nk}|$ does not depend on the order of summation (due to [Claim 3.9](#)). Thus, by [Def 9.10](#), $E = \bigcup_n E_n$ is a set of Lebesgue measure zero.

2) It should be clear that if $E = \{\bar{a}\} \subset \mathbb{R}^n$ is a one-point set, then, for every $\varepsilon > 0$, there exists a nondegenerate interval $I \subset \mathbb{R}^n$ such that $E = \{\bar{a}\} \subset I$ and $|I| < \varepsilon$. Indeed, if $\bar{a} = (a_1, a_2, \dots, a_n)$, then you can consider

$$I = [a_1 - \delta, a_1 + \delta] \times \cdots \times [a_n - \delta, a_n + \delta] \subset \mathbb{R}^n.$$

Then, by [Def 9.2](#), $|I| = (2\delta)^n$ and it is enough to set $\delta < \frac{1}{2} \sqrt[n]{\varepsilon}$. Since any finite (countable) set is a union of finitely (countably) many one-point sets, now [Item 2](#)) follows directly from [Item 1](#)).

3) The statement follows directly from [Def 9.10](#).

4) Such examples are relatively difficult to construct, thus we state this without proof. $\square$

Claim 9.2 (The Lebesgue Criterion for Riemann Integrability). Let

- $I \subset \mathbb{R}^n$ be a nondegenerate closed interval (see [Def 9.1](#));
- $f : I \rightarrow \mathbb{R}$ be a function on I ;
- $T(f) = \{\bar{x} \in I \mid f \text{ is discontinuous at } \bar{x}\}$ be the set of all points of discontinuity of the function f on the interval I .

Then, f is Riemann integrable on I (i.e. $f \in \mathcal{R}(I)$) if and only if f is bounded on I (i.e. $f \in \mathcal{B}(I)$) and the set $T(f)$ is of Lebesgue measure zero (i.e. $\mu(T(f)) = 0$). Thus,

$$(f \in \mathcal{R}(I)) \iff (f \in \mathcal{B}(I) \text{ and } \mu(T(f)) = 0).$$

Proof. Since the proof of Lebesgue's Criterion is very difficult (especially the $\Leftarrow$ direction), we state it without a proof. $\square$

Note that if some property holds at all points of a set X except possibly the points of a set of Lebesgue measure zero, then we say that the property holds *almost everywhere* on X . Thus, we can rephrase the Lebesgue Criterion like this: f is Riemann integrable on I iff f is bounded on I and f is almost everywhere continuous on I .

Let us consider another useful criterion for Riemann integrability.

Definition 9.11. Let

- I be an interval defined by Equality (1);
- P be a partition of the interval I defined as in Def 9.4;
- $f : I \rightarrow \mathbb{R}$ be a function on I ;
-

$$m_{i_1 i_2 \dots i_n} \stackrel{\text{def}}{=} \inf_{\bar{x} \in I_{i_1 i_2 \dots i_n}} f(\bar{x}), \quad M_{i_1 i_2 \dots i_n} \stackrel{\text{def}}{=} \sup_{\bar{x} \in I_{i_1 i_2 \dots i_n}} f(\bar{x}).$$

Then, the quantities

$$\underline{S}(f, P) \stackrel{\text{def}}{=} \sum_{i_1=1}^{t_1} \sum_{i_2=1}^{t_2} \cdots \sum_{i_n=1}^{t_n} m_{i_1 i_2 \dots i_n} |I_{i_1 i_2 \dots i_n}|$$

and

$$\overline{S}(f, P) \stackrel{\text{def}}{=} \sum_{i_1=1}^{t_1} \sum_{i_2=1}^{t_2} \cdots \sum_{i_n=1}^{t_n} M_{i_1 i_2 \dots i_n} |I_{i_1 i_2 \dots i_n}|$$

are called the *lower Darboux sum* and the *upper Darboux sum* (of the function f over the interval I corresponding to the partition P of the interval).

Claim 9.3 (Main Properties of Darboux Sums). Let

- I be an interval defined by Equality (1);
- $f : I \rightarrow \mathbb{R}$ be a function on I .

Then:

1. if a partition P' of the interval I is obtained by refining intervals of a partition P of I , then we have

$$\underline{S}(f, P) \leq \underline{S}(f, P') \leq \overline{S}(f, P') \leq \overline{S}(f, P);$$

2. for any two partitions P_1 and P_2 of the interval I we have

$$\underline{S}(f, P_1) \leq \overline{S}(f, P_2).$$

Proof. 1) Follows immediately from Def 9.11 (taking into account the definition of inf and sup).

2) Let P be an auxiliary partition of the interval I obtained by intersecting the intervals of the partitions P_1 and P_2 . For example:

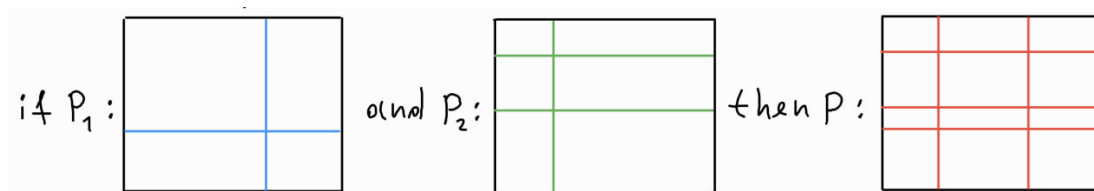


Figure 9.2

Since P is a refinement of each of the partitions P_1 and P_2 , by [Item 1](#)), we have

$$\underline{S}(f, P_1) \leq \underline{S}(f, P) \leq \overline{S}(f, P) \leq \overline{S}(f, P_2).$$

□

Definition 9.12. Let I be an interval defined by [Equality \(1\)](#) and let $f : I \rightarrow \mathbb{R}$ be a function on I . Then, the *lower Darboux integral* and the *upper Darboux integral* (of the function f over the interval I) are respectively defined to be:

$$\int_I f(\bar{x}) d\bar{x} \stackrel{\text{def}}{=} \underline{L}(f) \stackrel{\text{def}}{=} \sup_P \underline{S}(f, P),$$

and

$$\int_I f(\bar{x}) d\bar{x} \stackrel{\text{def}}{=} \overline{U}(f) \stackrel{\text{def}}{=} \inf_P \overline{S}(f, P),$$

where $\underline{S}(f, P)$ and $\overline{S}(f, P)$ are the lower and upper Darboux sums (see [Def 9.11](#)), and the supremum and infimum are taken over all possible partitions P of the interval I .

It should be clear that, due to [Def 9.11](#) and [Def 9.12](#), for any partition P of I , we have

$$\underline{S}(f, P) \leq \underline{L}(f) \leq \overline{U}(f) \leq \overline{S}(f, P).$$

Claim 9.4 (The Darboux Theorem). Let

- $I \subset \mathbb{R}^n$ be a nondegenerative closed interval (see [Def 9.1](#));
- $f : I \rightarrow \mathbb{R}$ be a bounded on I function.

Then:

1.

$$\lim_{\lambda(P) \rightarrow 0} \underline{S}(f, P) = \underline{L}(f);$$

2.

$$\lim_{\lambda(P) \rightarrow 0} \overline{S}(f, P) = \overline{U}(f).$$

Proof. 1) By the definition of the limit (see Calculus-1) we have:

$$\left(\lim_{\lambda(P) \rightarrow 0} \underline{S}(f, P) = \underline{L}(f) \right) \iff (\forall \varepsilon > 0 \exists \delta > 0 \forall P [\lambda(P) < \delta \Rightarrow |\underline{S}(f, P) - \underline{L}(f)| \leq \varepsilon]). \quad (3)$$

That is, it suffices to prove the right-hand statement. Fix $\varepsilon > 0$. Note that

$$(f \in \mathcal{B}(I)) \Rightarrow (\exists M > 0 \forall \bar{x} \in I [|f(\bar{x})| < M]).$$

Let

$$\widehat{\varepsilon} = \frac{\varepsilon}{2M + 1}.$$

By [Def 9.12](#) (essentially by the definition of the supremum), there exists a partition $\widehat{P}$ of the interval I such that

$$\underline{L}(f) - \widehat{\varepsilon} < \underline{S}(f, \widehat{P}). \quad (4)$$

Let $\widehat{T}$ be the set of all points of the interval I laying on the boundary of the intervals of the partition $\widehat{P}$. Taking into account the simple structure of $\widehat{T}$, it is not that difficult to see that there exists a number $\delta > 0$ such that the sum of the measures of those intervals that intersect $\widehat{T}$ is less than $\widehat{\varepsilon}$ for every partition P of I such that $\lambda(P) < \delta$. Indeed, take a look at the picture:

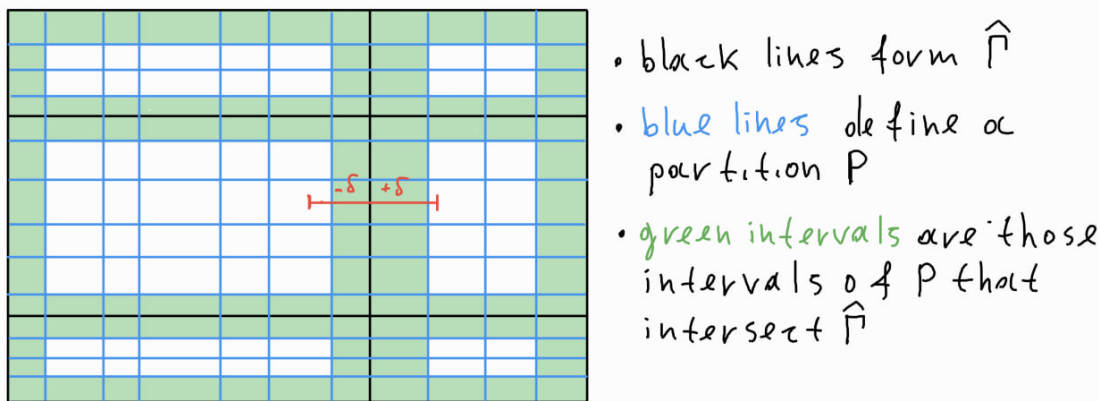


Figure 9.3

Since $\hat{\varepsilon}$ and $\hat{P}$ are fixed, it should be clear that there exists $\delta > 0$ such that the sum of the areas (2-measures) of the green intervals is less than $\hat{\varepsilon}$ for every partition P such that $\lambda(P) < \delta$.

Let P be any partition of I such that $\lambda(P) < \delta$. Due to (4), it remains to prove that

$$|\underline{S}(f, P) - \underline{L}(f)| \leq \varepsilon.$$

Let P' be an auxiliary partition of I obtained by intersecting the intervals of partitions $\hat{P}$ and P . Then, we have

$$\underline{L}(f) - \hat{\varepsilon} \stackrel{(4)}{<} \underline{S}(f, \hat{P}) \leq \underline{S}(f, P') \leq \underline{L}(f). \quad (5)$$

Also note that

$$\begin{aligned} |\underline{S}(f, P') - \underline{S}(f, P)| &\leq \sum_{\substack{I_i \in P \\ I_i \cap \hat{T} \neq \emptyset}} 2M |I_i| \\ &\leq 2M \hat{\varepsilon}. \end{aligned} \quad (6)$$

Indeed, the sums $\underline{S}(f, P')$ and $\underline{S}(f, P)$ both contain all the terms that correspond to intervals of the partition P that do not intersect $\hat{T}$, and if an interval I_i from P is such that $I_i \cap \hat{T} \neq \emptyset$, then, by the construction of a partition P' , there are intervals $I_{i1}, \dots, I_{ik}$ in P' such that their union is I_i . Then

$$\left| \sup_{I_i} f + \dots + \sup_{I_{ik}} f - \inf_{I_i} f \right| \leq 2 \sup_I f \leq 2M.$$

Since $\lambda(P) < \delta$, by the paragraph after Inequality (4), we have

$$\sum_{\substack{I_i \in P \\ I_i \cap \hat{T} \neq \emptyset}} |I_i| < \hat{\varepsilon}.$$

Therefore,

$$\begin{aligned} |\underline{S}(f, P) - \underline{L}(f)| &\leq |\underline{S}(f, P) - \underline{S}(f, P')| + |\underline{S}(f, P') - \underline{L}(f)| \\ &\leq 2M \hat{\varepsilon} + \hat{\varepsilon} = \varepsilon. \end{aligned}$$

Thus, we proved Item 1).

2) Can be proven similarly to Item 1). □

Claim 9.5 (Necessary Condition for Riemann Integrability). Let

- $I \subset \mathbb{R}^n$ be a nondegenerative closed interval (see Def 9.1);

- $f : I \rightarrow \mathbb{R}$ be a function on I .

Then,

$$f \in \mathcal{R}(I) \implies f \in \mathcal{B}(I).$$

That is, every Riemann integrable on I function is bounded on I .

Proof. Let f be unbounded on I . Then, for any partition P of the interval I , there exists an interval I_i of the partition P such that f is unbounded on I_i . If $(P, \bar{\zeta}')$ and $(P, \bar{\zeta}'')$ are two tagged partitions of P (see Def 9.8) with points $\bar{\zeta}'$ and $\bar{\zeta}''$ differing only in the choice of the points $\bar{\zeta}'_i$ and $\bar{\zeta}''_i$, then, by Def 9.8, we have

$$|\sigma(f, P, \bar{\zeta}') - \sigma(f, P, \bar{\zeta}'')| = |f(\bar{\zeta}'_i) - f(\bar{\zeta}''_i)| |I_i|. \quad (7)$$

Since $|I_i| > 0$ (indeed, by Def 9.4 all intervals of any partition of I are nondegenerate, thus, by Def 9.2, their n -measures are positive), by changing the points $\bar{\zeta}'_i, \bar{\zeta}''_i \in I_i$, as a result of unboundedness of f on I_i , we could make the right-hand side of Equality (7) arbitrarily large. Therefore, by the Cauchy criterion (see Calculus-1), this implies that the Riemann sums of f on I do not have a limit as $\lambda(P) \rightarrow 0$. Thus, by Def 9.9, f is not Riemann integrable on I .

It should be clear that $f \in \mathcal{B}(I) \not\Rightarrow f \in \mathcal{R}(I)$. $\square$

Claim 9.6 (The Darboux Criterion for Riemann Integrability). Let

- $I \subset \mathbb{R}^n$ be a nondegenerative closed interval (see Def 9.1);
- $f : I \rightarrow \mathbb{R}$ be a function on I .

Then, f is Riemann integrable on I if and only if it is bounded on I and its lower and upper Darboux integrals are equal. Thus,

$$(f \in \mathcal{R}(I)) \iff (f \in \mathcal{B}(I) \text{ and } \underline{L}(f) = \overline{U}(f)).$$

Proof. Necessity. If $f \in \mathcal{R}(I)$, then, by Claim 9.5, $f \in \mathcal{B}(I)$. Let

$$\int_I f(\bar{x}) d\bar{x} = s \in \mathbb{R}.$$

Then, by Def 9.9, we have

$$\forall \varepsilon > 0 \exists \delta(\varepsilon) > 0 \forall (P, \bar{\zeta}) [\lambda(P) < \delta \Rightarrow s - \frac{\varepsilon}{2} \leq \sigma(f, P, \bar{\zeta}) \leq s + \frac{\varepsilon}{2}]. \quad (8)$$

Since in (8) both inequalities hold true for every $\bar{\zeta}$, we have

$$\forall \varepsilon > 0 \exists \delta(\varepsilon) > 0 \forall P [\lambda(P) < \delta \Rightarrow s - \frac{\varepsilon}{2} \leq \inf_{\bar{\zeta}} \sigma(f, P, \bar{\zeta})], \quad (9)$$

and

$$\forall \varepsilon > 0 \exists \delta(\varepsilon) > 0 \forall P [\lambda(P) < \delta \Rightarrow \sup_{\bar{\zeta}} \sigma(f, P, \bar{\zeta}) \leq s + \frac{\varepsilon}{2}]. \quad (10)$$

Due to Def 9.11, it should be clear that

$$\inf_{\bar{\zeta}} \sigma(f, P, \bar{\zeta}) = \underline{S}(f, P), \quad \sup_{\bar{\zeta}} \sigma(f, P, \bar{\zeta}) = \overline{S}(f, P).$$

Therefore, (9) and (10) imply that

$$\forall \varepsilon > 0 \exists \delta(\varepsilon) > 0 \forall P [\lambda(P) < \delta \Rightarrow \overline{S}(f, P) - \underline{S}(f, P) \leq \varepsilon]. \quad (11)$$

Since, by Def 9.12, for every partition P we have

$$\underline{S}(f, P) \leq \underline{L}(f) \leq \overline{U}(f) \leq \overline{S}(f, P),$$

(11) implies that

$$\forall \varepsilon > 0 \quad 0 \leq \overline{U}(f) - \underline{L}(f) \leq \varepsilon.$$

Since the inequality $\overline{U}(f) - \underline{L}(f) \leq \varepsilon$ does not depend on P , this is obviously equivalent to

$$\underline{L}(f) = \overline{U}(f).$$

Sufficiency. Note that, due to [Def 9.11](#) and [Def 9.12](#), we have

$$\forall (P, \bar{\zeta}) \quad \underline{S}(f, P) \leq \sigma(f, P, \bar{\zeta}) \leq \overline{S}(f, P). \quad (12)$$

Since, by [Claim 9.4](#),

$$\lim_{\lambda(P) \rightarrow 0} \underline{S}(f, P) = \underline{L}(f) \quad \text{and} \quad \lim_{\lambda(P) \rightarrow 0} \overline{S}(f, P) = \overline{U}(f),$$

and it is given that $\underline{L}(f) = \overline{U}(f) = s$ for some $s \in \mathbb{R}$, by the Squeeze theorem (see Calculus-1), from [Inequality \(12\)](#) we have

$$\lim_{\lambda(P) \rightarrow 0} \underline{S}(f, P) = \underline{L}(f) = s \leq \lim_{\lambda(P) \rightarrow 0} \sigma(f, P, \bar{\zeta}) \leq s = \overline{U}(f) = \lim_{\lambda(P) \rightarrow 0} \overline{S}(f, P). \quad (13)$$

That is, due to [Def 9.9](#), $f \in \mathcal{R}(I)$.

Note that, if $f \in \mathcal{R}(I)$, then

$$\underline{L}(f) = \overline{U}(f) = \int_I f(\bar{x}) d\bar{x}.$$

□